DFT reference vs MLIP prediction | train context, held-out test, and count density

A Scratch-600K
Spearman: 0.45
Pearson: 0.46
$E_0 = -2195$ eV

B FT-600K
Spearman: 0.47
Pearson: 0.44
$E_0 = -2195$ eV

C Scratch-MultiT
Spearman: 0.87
Pearson: 1.00
$E_0 = -2193$ eV

D FT-MultiT
Spearman: 0.86
Pearson: 1.00
$E_0 = -2193$ eV

MLIP energy / atom (eV)

DFT energy / atom (eV)

Count per hexbin

E Scratch-600K
Spearman: 0.84
Pearson: 0.80

F FT-600K
Spearman: 0.88
Pearson: 0.88

G Scratch-MultiT
Spearman: 0.86
Pearson: 0.74

H FT-MultiT
Spearman: 0.90
Pearson: 0.90

MLIP force component (eV Å^{-1})

DFT force component (eV Å^{-1})

train held-out test

Count per hexbin